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Global momentum: The optimal trading approach

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ABSTRACT

We investigate momentum strategies in international equity markets. International investors that switch back and forth from one country to the other based on their previous performances can earn more than 2.53% percent per month or 35% per year and momentum effect is substantially strong in emerging markets with returns up to 2.41% per month or 33% per annum. For the international investor, we identify world risk factors, to our knowledge, first time in the literature. We find that higher profits for international momentum portfolios are mainly due to predictability from world macroeconomic risk factors. Notably, the results confirm the informational role of world industrial production.

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1. Introduction

The observation that stocks that exhibited positive performance in the past (winners) are likely to continue to perform well in the future (3–12 months) and these stocks tended to outperform previous losers has been one of the most celebrated and persistent market anomalies. Three decades after its initial discovery, reports persist in favour of the profitability of momentum trading strategies. This anomaly has puzzled analysts, market participants, academics, and a wide range of stakeholders and goes against the established notion that the information is immediately incorporated into prices. Instead, the information is gradually absorbed and reflected in the stock prices (Frazzini, 2006; Otchere and Chan, 2003; Savor, 2012). Although the momentum idea was initially confined to US Stock Markets (Jegadeesh and Titman, 1993, 2001), the momentum strategy has gained international support (Goyal and Wahal, 2015; Griffin et al., 2003; Rouwenhorst, 1998) and is evidenced across asset classes. Indeed, the empirical findings show that momentum strategies have been yielding positive returns since 1985 for nearly all equity indices, fixed income futures, commodity futures, and currency forwards (see, for example, Asness et al., 2013; Filipou et al., 2018; Fung and Hsieh, 1997, 2001; Georgopoulou and Wang, 2016; Grobys et al., 2018; Hutchinson and O'Brien, 2020; Liu et al., 2021; Menkhoff et al., 2012; Moskowitz et al., 2012), more recently in cryptocurrency markets (Zaremba et al., 2021; Li et al., 2021) and at the intraday frequency (Gao et al., 2019). Momentum's presence has only been challenged in a limited number of markets where momentum profits are not present in Southeast Asian markets (see Jegadeesh and Titman, 2001), especially in China (Gao et al., 2021) and Latin America (Berggrun et al., 2020). In any case, momentum is regarded as one of the few robust market anomalies and poses an explanatory problem on the Capital Asset Pricing Model (CAPM) as well as on the Fama and French (1993, 2015) three-and five-factor models. There have been numerous attempts to explain the persistence of the momentum effect and the explanations tend to revolve around industrial structure and firm-specific indicators (Conrad and Gautam, 1998; Gu and Wu, 2020; Li et al., 2009; Walkshäusl, 2019). Another stream of research attempts to position momentum on the top of the normative and cognitive pillars as exhibited by the behaviour of the participants in the financial markets and open up the question of whether the momentum results are also affected by cultural dimensions when it comes to investing in the international financial markets (see Barberis et al., 1998; Chen and Lu, 2017; Dierick et al., 2019).

All the empirical studies rely on some sort of publicly available information to make their arguments and our paper does not deviate from the previous norm. The spirit of our study is the same but we take a different approach in order to shed some further light on the momentum puzzle. The contribution of the paper to the relevant literature is as follows: *First*, we take into

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consideration the profile of investment on a global scale. Specifically, in this paper, we investigate a global momentum strategy where the international investor selects indices from the world equity markets by implementing global momentum strategies. As an ever-increasing number of international equity funds gain access to foreign equity markets due to globalization, portfolio managers have to make top-down decisions when it comes to international asset allocations. Investors tend to assume global investment strategies in the light of risk diversification amongst equity markets, induced by the ever-increasing degree of global integration and trade easiness. For example, Exchange Traded Funds (ETFs) provide investors with the opportunity to invest worldwide.

Previous relevant research investigating momentum strategies (see Barroso and Santa-Clara, 2016; Bohl et al., 2016; Chen et al., 2017) somehow failed to look into a specific momentum strategy that puts a typical international investor at his helm switching from one equity market to another. To the best of our knowledge and for the first time in the literature, we introduce measures for global risks faced by international investors by focusing on country indices. Scarce previous evidence on multiple equity indices focused on stocks for selected indices. For example, Rouwenhorst's (1998) empirical study provides evidence on momentum strategies for equities in 12 European markets and Rouwenhorst (1999) examines whether momentum effects are present among stocks listed in emerging markets. Although both studies find that momentum strategies earn high abnormal returns in developed and emerging equity markets, they use stocks on internationally diversified portfolios rather than indices as we propose in this study. Griffin et al. (2005) report significant momentum returns in a variety of markets across the world from 1975 to 1995 but focus on stocks rather than indices from a US investor viewpoint. Nevertheless, the winning portfolios in Griffin et al. (2005) study tend to outperform the corresponding indices in America, Europe, and Asia. Chan et al. (2000) focuses on international indices but include exchange rate movements as they approach the momentum strategies from the viewpoint of a US investor interested in US Dollars returns. Chan et al. (2000) construct portfolios based on indices weighted by their US Dollar returns minus the average US Dollar return of the "world market" of 23 indices from 1980 to 1995. A country index with a higher return than the average of the "world market" will get a higher weight compared to a country index with a lower return than the "world market". Hence, in Chan et al. (2000), the amount invested in one country index depends on how much the country index outperformed the "world market" of the 23 indices all denominated in US Dollars. Chan et al. (2000) also report significant momentum profits for short holding periods (less than 4 weeks). Our study deviates from weighted index-based portfolios and considers the indices themselves. Each of these countries' indices is a well-diversified basket of stocks and all indices are divided into deciles based on their returns over different formation periods with a view to implementing zero-cost portfolios. By extending the established analysis to a global equity market setting, we assess whether intermediate past performance offers a more beneficial information for internationally aligned investors. This is also important because investment managers operating globally must make top-down decisions on international asset allocation.

Secondly, we examine the momentum portfolios with an array of formation and holding periods across the world with a view to providing a compelling answer to the question *"Is there any optimal formation/holding period?"* and we experiment with various formations and holding periods as suggested by the empirical literature. Our results indicate that there is an optimal period of 9-month/3-month (formation/holding) period. We start with 47 global equity markets from 1969 to 2020 and then we move away from the full sample and we delve into established and developing markets, different time spans-all tied up with a variety of formation and holding periods. Baker et al. (2016) argue that the market uncertainty across the world has intensified and our approach aspires to take into consideration certain events and to examine empirically whether the momentum strategy is upheld given the different economic circumstances. Hence, we perform a comparative study among equities with different characteristics and equity market development from different geographical regions and extend the results of previous studies (which examined country-specific indices) by adopting a more consolidated approach.

Thirdly, we perform a comparative study on the impact of the market on momentum strategies. Besides the relatively high profitability, momentum has occasionally experienced persistent strings of negative returns. In 1932, the momentum factor for the U.S. equity market exhibited a crash of −67.10%. Also in 2009, the momentum factor for both the U.S. and international (ex-U.S.) equity markets recorded a crash of −73.42% in three months (see Barroso and Santa-Clara, 2016; Daniel and Moskowitz, 2016). Daniel and Moskowitz (2016) show that when market conditions improve following bear markets, those losers make strong gains which, in turn, result in a "momentum crash". We do not try to investigate the impact and potential predictability of these momentum crashes, which appear to be a key and robust feature of momentum strategies but armed with the optimal formation/holding period and using the MSCI index from 1969 to 2020, we try to assess the performance of the momentum strategies conditional upon bear and bull market phases.

Fourthly, institutional investors build their asset allocation strategies while referring to a multi-factor portfolio exposed to various risks. Following this framework (see, for example, Jegadeesh and Titman (1993)), we examine for the first time the role of world risks factors in explaining the global momentum profits and find that the global momentum strategies yield significant abnormal returns after adjusting consecutively for world idiosyncratic risks and world market state risk. More importantly, we establish a strong link between world macroeconomic risk factors and the global momentum, notably the world industrial production. Our findings contrast with Asness et al. (2013), Chordia and Shivakumar (2006), Griffin et al. (2003), Liew and Vassalou (2000) and Markus and Harald (2012) who failed to establish a link between price momentum and macroeconomic risk factors. However, we concede that their analysis was conducted on individual countries while we are considering the global momentum as a global coordinated and generalized phenomenon. In addition, the previous studies mainly focus on GDP as the major macroeconomic risk factor while certain macroeconomic risk factors were overlooked. Our analysis offers a more consolidated approach to price momentum and macroeconomic risk factors by considering more of the latter.

Overall, our results uniquely position the momentum at the international level by investing in country indices worldwide, reflecting thereby the notion of international integration where the world is treated as an investment playground. By focusing on indices, we introduce measures for global risks facing international investors, and to the best of our knowledge, for the first time in the literature. We show that switching back and forth investments from one country to the other, picking winners and selling losers yields almost 2.53% per month, about 35% per annum almost two to three times the previously reported momentum returns. We included forty-seven (47) global equity markets both developed and emerging, a variety of formation and holding periods, and sub-period analysis to cover various world events. Based on the market phases (bull versus bear) and utilizing the optimal formation/holding period, (9 months/3 months)

our results identify positive returns following both bull and bear phases with the former being more pronounced. These findings diverge from past examples by showing that momentum-based trading strategies in a global equity market setting generate positive returns even during the *bear* market phases from 1969 to 2020 based on intermediate past performance. This appears to deepen the puzzle of momentum. Furthermore, our findings also demonstrate that the high returns to the international momentum investor are compensation for world risk factors and endorse the informational role of industrial production as a risk factor that must be taken into account. Explicitly, global momentum strategies obtain significant abnormal returns after adjusting for world idiosyncratic risks. We also report a strong and positive relation between world industrial production and momentum returns indicating international investors might be able to earn extra returns by increasing their exposure to world risk factors. Hence, our findings reinforce the inclusion of momentum as a factor in linear factor models (Carhart, 1997); shed light on the impact of arbitrage trading (Chen and Zhang, 2010; Daniel and Titman, 1997), and signposts that the momentum premium occurs during sub-periods and market states.

The rest of our paper is set out as follows. Section 2 presents the empirical framework of analysis of the Global Momentum Strategies. In Section 3 we present the data. Section 4 discusses the empirical result, and Section 5 concludes the paper.

2. Empirical framework

This study follows the methodological framework set out by Jegadeesh and Titman (1993, 2001)). Nevertheless, this study does not focus on stocks, but it employs global equity indices from the viewpoint of an international investor who has the capacity of switching between worldwide equity indices using a momentum trading strategy. This study assumes no trading restrictions and zero-cost trading and delves into non-overlapping and overlapping portfolios. With reference to *non-overlapping* portfolios, the momentum strategy selects indices based on returns over the past J-months and holds them for K-months (we will refer to this as a J-month/K-month strategy). The empirical investigation begins with ranking the indices based on their past J-month returns (where J = 3, 6, 9, 12 months), and based on these rankings, ten decile portfolios are formed that equally weight the indices contained in the top decile, the second decile, and so on. The top decile portfolio and the bottom decile portfolio are called "losers" and "winners" respectively. At the end of each J-month formation period, international investors buy the winner portfolio and sell the loser portfolio, holding this position for K-months (where K = 3, 6, 9, or 12 months). This yields a total of 16 strategies (see Table 2). To increase the power of our tests, this study also examines a second set of 16 strategies involving *overlapping* portfolios where the strategy in any given month buys the winner portfolio and sells the loser portfolio, holding this position for K months, The strategy holds a series of portfolios that are selected in the current month as well as in the previous months of the holding period, K (see Table 5).

The equally-weighted average return for each portfolio is computed as:

$$R_W = \frac{1}{Np}\sum_{n=1}^{N_p}\left[\sum_{t=1}^{k} r_{i,t}^{w}\right] \tag{1}$$

$$R_L = \frac{1}{N_p}\sum_{n=1}^{N_p}\left[\sum_{t=1}^{k} r_{i,t}^{L}\right] \tag{2}$$

where k is the holding period, R_L represents the loser portfolio average return in t month, R_W represents the winner portfolio average return in t month, and N_p, is the number of losers or the winners in the portfolios. This analysis is performed N times and we compute the cumulative average return of the losers' and the winners' portfolios as follows.

$$AR_w = \frac{1}{N}\sum_{n=1}^{N} R_w \tag{3}$$

$$AR_L = \frac{1}{N}\sum_{n=1}^{N} R_L \tag{4}$$

The next step is to compute the momentum strategies, which can be defined as a strategy of buying indices that have performed well in the past and selling indices that have performed badly in the past. The returns are computed as follows.

$$Mom = AR_W - AR_L \tag{5}$$

For the momentum strategies, the T statistics can be computed as follows:

$$H_0 : (AR_W - AR_L) > 0$$

$$T_{M,t} = (AR_W - AR_L) / \sqrt{2S_{C,t}^2/N} \tag{6}$$

where the variance of the difference of two samples of equal size N equal to $2S_{C,t}^2/N$ and $S_{M,t}^2$ is computed as: $S_{M,t}^2 = \left[\sum_{n=1}^{N}(R_W - AR_W)^2 - \sum_{n=1}^{N}(R_L - AR_L-)^2\right]/2(N-1)$.

We test whether global investors earn significant returns after adjusting for various risk factors including idiosyncratic risks, market state risks, and macroeconomic risks. For all estimations, we use the Generalized Method Moments-GMM. Zivot and Wang (2006) argue that this method accounts for possible endogeneity problems and does not require us to make distributional assumptions as the coefficients are derived using the maximum likelihood estimation. For robustness, we also estimated the same models using OLS and the Newey–West procedure, and the results remain unchanged.

To estimate momentum returns, we first side with Fama and French's (1993) three-factor model and account for idiosyncratic risk factors as follows:

$$Mom_t = \alpha_0 + \beta_1 (R_M - R_F)_t + \beta_2 SMB_t + \beta_3 HML_t + \varepsilon_t \tag{7}$$

where Mom_t represents the returns of the momentum portfolios at time t, $R_M - R_F$ is the US excess market return, HML_t (High minus Low) is the return to a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks in the US, and SMB_t (Small minus Big) is a return to long-short portfolios constructed using the size in US market. α_0, β_1, β_2, and β_3 are coefficients to be estimated. We extend the analysis of idiosyncratic risk by including two additional factors: Robust-minus-Weak (RMW) profitability and Conservative-minus-Aggressive (CMA) investment factors to form the so-called Fama and French 5-factor model (2015). Variables' definitions are discussed in Appendix A.

$$Mom_t = \alpha_0 + \beta_1 (R_M - R_F)_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + \varepsilon_t \tag{7a}$$

Next, we regress global momentum portfolios' returns on the following world market state factors that we define for international investors:

$$Mom_t = \alpha_1 + \gamma_1 (R_M - R_F)_t + \gamma_2 SMB_t + \gamma_3 HML_t + \gamma_4 RMW_t + \gamma_5 CMA_t + \gamma_6 LIQ_{t-1} + \gamma_7 DS_{t-1} + \gamma_8 TS_{t-1} + \gamma_9 MKT_{t-1} + \varepsilon_t \tag{8}$$

where LIQ_{t-1} is the Pastor and Stambaugh (2003) Level of Aggregate Liquidity.[1] DS_{t-1} is the Default Spread, TS_{t-1} is the Term Spread, and MKT_{t-1} is the MSCI world indices price level, all lagged one period to avoid spurious regression and for the purpose of avoiding potential problems that might arise as returns from the momentum portfolio and the independent variables are probably jointly determined, α_1, γ_1, γ_2, γ_3, and γ_4 are coefficients to be estimated. Variables definitions are discussed in Appendix B.

Finally, we introduce the following macroeconomic world risk factors as presented in Eq. (9) below:

$$Mom_t = \alpha_1 + \delta_1 (R_M - R_F)_t + \delta_2 SMB_t + \delta_3 HML_t + \delta_4 RMW_t + \delta_5 CMA_t + \delta_6 \Delta OP_{t-1} + \delta_7 WVOL_{t-1} + \delta_8 \Delta IP_{t-1} + \varepsilon_t \quad (9)$$

where ΔOP_{t-1} is the percent change in monthly oil price, $WVOL_{t-1}$ is the world market volatility (measured as the standard deviation of the return on the MSCI world indices), ΔIP_{t-1} is the (percent change) in the monthly value of the US Industrial production that we use as a proxy for world production. α_2, δ_1, δ_2, and δ_3 are coefficients to be estimated. The explanatory variables lagged one period to avoid spurious regressions and endogeneity. Variables definitions are discussed in Appendix C

We further examine the simultaneous impact of the idiosyncratic risks, markets state risks, and macroeconomic risks on world momentum's returns using the following models initiated by Eqs. (7) and (7a) respectively:

$$Mom_t = \alpha_3 + \theta_1 (R_M - R_F)_t + \theta_2 SMB_t + \theta_3 HML_t + \theta_4 \Delta LIQ_{t-1} + \theta_5 DS_{t-1} + \theta_6 TS_{t-1} + \theta_7 MKT_{t-1} + \theta_8 \Delta OP_{t-1} + \theta_9 WVOL_{t-1} + \theta_{10} IP_{t-1} + \varepsilon_t \quad (10)$$

$$Mom_t = \alpha_3 + \theta_1 (R_M - R_F)_t + \theta_2 SMB_t + \theta_3 HML_t + \theta_4 RMW_t + \theta_5 CMA_t + \theta_6 \Delta LIQ_{t-1} + \theta_7 DS_{t-1} + \theta_8 TS_{t-1} + \theta_9 MKT_{t-1} + \theta_{10} \Delta OP_{t-1} + \theta_{11} WVOL_{t-1} + \theta_{12} IP_{t-1} + \varepsilon_t \quad (10a)$$

where all variables are defined as above and in Appendices A–C. We test for the null hypothesis that the estimated intercepts (alphas) are significantly different from zero. If the alphas are statistically different from zero, we conclude that the global risk factors do not explain world momentum profits fully and there are still some unobserved factors that are captured by the statistical significance of the intercept. On the other hand, if the alphas are zero then we conclude momentum profits are compensation for the related risks bared by the investor. In other words, the set of independent variables carries out all the explanatory power of the model.

We extend our analysis to test the momentum strategies in emerging markets, developed markets, and established markets separately. We further outspread our analysis to the globalization periods (1994–2020), the financial crisis periods, and business cycle fluctuations (seven expansions and seven contractions according to NBER business cycles specifications) as part of the quest to assess the pervasiveness of the momentum profits. The global momentum trading strategy is applied in the full sample to all the 47 countries (1969–2020) and subsamples of countries that have stock markets from the beginning of our research period in 1969 to 2020 (18 countries) and subsamples of developed (23 indices starting December 1969) versus emerging countries (24 indices starting December 1987).

[1] We have alternative estimations using EURBOR_OIS which is considered a market barometer of liquidity and widely used as a measure of the level of liquidity in the European region. Conclusions do not change. Results are available from the authors.

3. Data

The data set includes equity market indices from forty-seven world markets; twenty-three of them are developed markets and twenty-four are emerging markets. The data period covers December 1969 through February 2020. We use constituents of the MSCI world index denominated in US dollars and this study is conducted from the viewpoint of US-based investors. These forty-seven indices are selected based on their tradability and international investors can mimic these country indices with a limited number of stocks (see, for example, Blitz and Hiuj, 2012; Fisher and Lorie, 1970; Frino and Gallagher, 2001; Miziołek and Feder-Sempach, 2019). The first monthly returns are calculated in January 1970 for the countries trading stocks at that time and continued trading throughout the research period (USA, Japan, UK, Australia, France, Germany, Italy, Canada, Hong Kong, Singapore, Spain, Switzerland, Belgium, Sweden, Austria, Netherlands, Norway, and Denmark). New Zealand and Finland enter the sample in December 1981. Ireland and Portugal together with eleven emerging countries (Brazil, Korea, Turkey, Indonesia, Mexico, Taiwan, Thailand, Argentina, Malaysia, Chile, and Jordan) enter in December 1987. Later on, Israel and further eight emerging countries (China, India, South Africa, Colombia, Poland, Pakistan, Sri Lanka, and Peru) join in December 1992. Finally, in December 1994 we add five more emerging countries (Russia, Egypt, Czech Rep, Hungary, and Morocco).

Table 1 presents descriptive statistics. The mean returns are close to zero as expected but the standard deviation varies amongst countries with emerging markets exhibiting higher volatility (up to 0.15) than their developed counterparts (up to 0.09). Based on the values of skewness and kurtosis as well as the values of Shapiro–Wilk (SW) tests, the returns seem not to follow the normal distribution. The latter provides further justification for using the GMM methodology.

We classify world risk factors into three different groups, namely, idiosyncratic risks, market state risks, and macroeconomic risks (see Appendices A, B, and C for detailed definition). The world risk factors are not readily available, but we get around this problem for idiosyncratic and market risks by following Chako and Evans (2014), and we utilize the US risk factors as proxies. Chako and Evans (2014) justify this approximation by arguing that the global diversity of assets held by U.S. companies makes the U.S. equity market a fair proxy of world asset markets. Haslam et al. (2013) provide also compelling evidence regarding the dominance of United States-based companies on the world equity markets and treating the Unites States markets as a proxy for the world volatility and trends.

To estimate liquidity, we follow the methodology described by Pastor and Stambaugh (2003). For the monthly aggregate liquidity measure, we calculate a cross-sectional average of individual-stock liquidity measures in the United States. Each stock's liquidity in a given month is estimated by using that stock's within-month daily returns and volume.

Default spread is calculated as the difference between the average yields of US bonds rated BAA by Moody's and the average yields of bonds with Moody's rating of AAA (see Appendix B for detailed definition). The term spread is calculated as the difference between the average yield of the US Treasury bonds with more than 20 years to maturity and the average yield of the T-bill that matures in three months (for a detailed definition see Appendix B). These rates are collected from WRDS.

We use the world industrial production index level (Total Industrial Production and Manufacturing Production Index value SDDS+) from the Federal Reserve database. We calculate the Monthly differences in the natural logarithm of the industrial production indexes $\Delta IP = \ln(IP_t) - \ln(IP_{t-1})$, at time t and t−1.

Table 1
Monthly returns: 1969–2020.

Panel A: Monthly returns — Developed countries								
Name ID	Start	End	Mean	Std Dev	Skewness	Kurtosis	Shapiro Wilk (SW)	SW Test of significance
USA (1)	31/12/1969	28/01/2020	0.01	0.04	−0.67	2.45	0.97	0.00
Japan (2)	31/12/1969	28/01/2020	0.01	0.06	−0.02	0.87	0.99	0.00
UK (4)	31/12/1969	28/01/2020	0.00	0.06	0.26	5.57	0.94	0.00
Australia (10)	31/12/1969	28/01/2020	0.00	0.07	−1.49	9.75	0.92	0.00
France (11)	31/12/1969	28/01/2020	0.01	0.06	−0.44	1.65	0.97	0.00
Germany (12)	31/12/1969	28/01/2020	0.01	0.06	−0.62	1.87	0.97	0.00
Italy (15)	31/12/1969	28/01/2020	0.00	0.07	−0.17	0.87	0.99	0.00
Canada (20)	31/12/1969	28/01/2020	0.00	0.06	−0.60	3.52	0.96	0.00
Hong Kong (21)	31/12/1969	28/01/2020	0.01	0.09	−0.53	7.85	0.91	0.00
Singapore (23)	31/12/1969	28/01/2020	0.01	0.08	−0.51	6.17	0.92	0.00
Spain (24)	31/12/1969	28/01/2020	0.00	0.07	−0.46	2.39	0.97	0.00
Switzerland (25)	31/12/1969	28/01/2020	0.01	0.05	−0.39	1.44	0.98	0.00
Belgium (27)	31/12/1969	28/01/2020	0.00	0.06	−1.01	7.39	0.92	0.00
Sweden (29)	31/12/1969	28/01/2020	0.01	0.07	−0.47	1.50	0.98	0.00
Austria (30)	31/12/1969	28/01/2020	0.00	0.07	−0.95	6.62	0.92	0.00
Ireland (32)	31/12/1987	28/01/2020	0.00	0.06	−0.91	2.71	0.95[1,2 and 3]	0.00
Netherlands (33)	31/12/1969	28/01/2020	0.01	0.06	−0.78	2.72	0.96	0.00
New Zealand (34)	31/12/1981	28/01/2020	0.00	0.07	−0.84	4.94	0.95	0.00
Norway (35)	31/12/1969	28/01/2020	0.01	0.08	−0.78	2.97	0.96	0.00
Portugal (37)	31/12/1987	28/01/2020	0.00	0.07	−0.37	1.66	0.98	0.00
Denmark (39)	31/12/1969	28/01/2020	0.01	0.06	−0.44	2.32	0.97	0.00
Finland (40)	31/12/1981	28/01/2020	0.01	0.08	−0.38	2.06	0.97	0.00
Israel (42)	31/12/1992	28/01/2020	0.00	0.07	−0.49	0.95	0.97	0.00
Panel B: Monthly returns — Emerging markets								
China (3)	31/12/1992	28/01/2020	0.01	0.09	−0.07	2.03	0.96	0.00
Brazil (5)	31/05/1987	28/01/2020	0.01	0.14	−1.32	10.85	0.90	0.00
India (6)	31/12/1992	28/01/2020	0.01	0.08	−0.34	1.27	0.98	0.00
Korea (7)	31/12/1987	28/01/2020	0.00	0.10	0.19	3.61	0.96	0.00
Russia (8)	30/12/1994	28/01/2020	0.01	0.14	−1.20	7.33	0.91	0.00
Turkey (9)	31/12/1987	28/01/2020	0.00	0.15	−0.02	1.46	0.98	0.00
Indonesia (13)	31/12/1987	28/01/2020	0.01	0.12	0.14	6.27	0.90	0.00
South Africa (14)	31/12/1992	28/01/2020	0.00	0.08	−0.87	2.34	0.96	0.00
Mexico (16)	31/12/1987	28/01/2020	0.01	0.09	−1.01	3.87	0.94	0.00
Taiwan (17)	31/12/1987	28/01/2020	0.00	0.09	−0.10	2.28	0.97	0.00
Thailand (18)	31/12/1987	28/01/2020	0.00	0.10	−0.54	3.13	0.94	0.00
Argentina (19)	31/12/1987	28/01/2020	0.01	0.14	−0.10	4.62	0.93	0.00
Malaysia (22)	31/12/1987	28/01/2020	0.00	0.07	−0.25	5.44	0.92	0.00
Chile (26)	31/12/1987	28/01/2020	0.01	0.07	−0.40	1.98	0.98	0.00
Colombia (28)	31/12/1992	28/01/2020	0.00	0.09	−0.81	3.72	0.96	0.00
Egypt (31)	31/05/1994	28/01/2020	0.01	0.09	−0.44	2.64	0.97	0.00
Poland (36)	31/12/1992	28/01/2020	0.00	0.12	0.55	6.63	0.93	0.00
Czechia (38)	31/12/1994	28/01/2020	0.00	0.08	−0.75	2.78	0.96	0.00
Hungary (41)	30/12/1994	28/01/2020	0.01	0.10	−1.06	4.85	0.93	0.00
Pakistan (43)	31/12/1992	28/01/2020	0.00	0.10	−1.28	8.02	0.90	0.00
Sri Lanka (44)	31/12/1992	28/01/2020	0.00	0.09	0.52	3.90	0.94	0.00
Morocco (45)	31/12/1994	28/01/2020	0.00	0.05	−0.54	3.44	0.96	0.00
Peru (46)	31/12/1992	28/01/2020	0.01	0.09	−0.72	3.64	0.95	0.00
Jordan (47)	31/12/1987	28/01/2020	0.00	0.06	−1.59	12.6	089	0.00

This table reports the descriptive statistic and normality test of individual countries. The sample is from 1969 to 2020. We test whether the returns of the 23 developed and 24 Emerging markets indices prices are normally distributed through their skewness, kurtosis. We use the mean and standard deviation to compare the two sets of countries. We used the Shapiro-Wilk test to confirm the normality of the distribution. The results are ** statistically significant for $p > 0.05$ and ***statistically significant for $p > 0.1$.

The Oil Price as a series (Producer price index crude petroleum series) was obtained from the Bureau of Labour Statistics database. We calculate Monthly differences in the natural logarithm of the producer price index crude petroleum series (obtain from the Bureau of Labour Statistics, U.S. Department of Labour) $\Delta OP = \ln(OPt) - \ln(OPt-1)$ at time t and t−1.

4. Empirical results

4.1. Returns to global momentum strategies

We commence our investigation by implementing the basic momentum strategies on the full sample of forty-seven countries during the period from 1969 to 2020 (see Table 2). In contrast to previous studies that focus on buying and selling individual stocks in specific markets, our analysis focuses on indices for the reasons that we described in our Introduction. The first set of our findings is shown in Table 2 which indicates high momentum profits for international investors. In general, momentum strategies, irrespective of the formation and holding periods, are profitable at the international level: the top (or winner) portfolios outshine the bottom (or loser) portfolios. Remarkably, Table 2 shows that all the momentum strategies' returns are significant and positive. The only period which is associated with negative returns is when the formation takes 12 months, and the holding period is 3 and 6 months respectively but the returns fail to attain any conventional level of significance. The most successful momentum strategy selects indices based on their previous performances over 9 months and then holds the portfolio for the next 3 months and yields statistically and economically significant momentum returns of approximately 2.53% per month (35% per annum). Our results confirm that momentum returns tend to go against the capital market efficiency in international equity markets confirming the risk-based or behaviour-based explanations

Table 2
Momentum returns for international investor non-overlapping portfolio: Full sample 1969–2020.

J	K=	3	6	9	12
3 Sell		0.36	0.45	0.31	0.40
3 Buy		0.77	0.79	0.90	0.78
3 Buy-Sell		0.40	0.34	0.59***	0.38**
		(1.17)	(1.44)	(3.06)	(2.36)
Sig		0.24	0.15	0.00	0.01
SR*		0.01	−0.01	0.08	0.00
6 Sell			0.30	0.14	0.44
6 Buy		0.87	0.93	0.84	0.67
6 Buy-Sell		0.44	0.62*	0.69**	0.23
		(0.91)	(1.92)	(2.54)	(1.03)
Sig		0.37	0.05	0.01	0.31
SR		0.01	0.08	0.12	−0.07
9 Sell		−0.77	−0.00	0.12	0.19
9 Buy		1.76	0.87	0.99	0.85
9 Buy-sell		2.53***	0.90**	0.88**	0.66**
		(4.20)	(2.35)	(2.58)	(2.13)
Sig		000	0.02	0.01	0.04
SR		0.44	0.17	0.18	0.11
12 Sell		1.27	0.82	0.21	0.45
12 Buy		1.03	0.61	0.38	0.50
12 Buy-Sell		−0.24	−0.22	0.18	0.05
		(−0.34)	(−0.44)	(0.46)	(0.14)
Sig		0.74	0.66	0.65	0.89
SR		−0.12	−0.17	−0.08	−0.12

This table reports the momentum strategies' returns implemented on 47 stock market indices. The sample is from 1969 to 2020. I form momentum portfolio (buy past winners and sell past losers) based on past performance of stock market indices. I use 4 different formations and holding periods. The winner portfolios, the loser portfolios, and the winner minus loser portfolios monthly return are reported. The t-statistic are reported in the brackets and the results are statistically significant for $p < 0.05$ and the Sharpe Ratio (SR) for the full sample 1969–2020 is reported.
*: SR — Sharpe Ratio.

put forward by the empirical literature to account for this pattern. Furthermore, Table 2 reports relatively low Sharpe Ratios (SR) pointing to the volatility ingrained in momentum strategy over time. This is in line with Barroso and Santa-Clara (2016) who suggest that the risk of momentum is highly variable over time but predictable.

Given the heterogeneity of the full sample of forty-seven markets and the start of the indices at different points in time, Table 3 makes an explicit attempt to account for this type of heterogeneity by delving deeper into momentum. Table 3 splits the sampled indices into established markets (18 indices) versus the full sample of 47 indices with all the indices starting in 1994 creating thereby two balanced panels where all the markets enjoy a common starting point. Regarding the established markets, the best momentum strategy significantly yields 1.68% per month (21.1% per annum) with the 9-month formation and 3-month holding period which coincides with the results in Table 2. Turning our sights on the full sample of 47 countries where the markets provide us with data and coincide with the onset of the Globalization period in 1994, (hence the title "Globalisation") the results remain the same: with all the indices examined, momentum strategies are still profitable, and most importantly the optimal period still remains the same: 9-month formation followed by a 3-month holding period. Remarkably, the returns appear to be higher when the entire sample is taken into consideration: As it is evident when the period from 1994 onwards is considered and with all the markets on board, the momentum strategy yields 2.10% return per month (28.3% per annum) as opposed to 1.68% (21.1% per annum) when only the established markets are considered. These results appear to be in line with the findings by Griffin et al. (2005) which reported significant momentum results for the 6-month/6-month period although they employ only the aforementioned strategy and of course, used a considerably different time period.

Table 4 sheds further light on the momentum strategies by considering the developed markets (1969–2020) vis-à-vis emerging ones (1988–2020). The differences in the starting period are dictated by the data availability. The momentum effect persists and the momentum strategies appear to yield positive returns for the investors adding to the robustness of the previous results. Specifically, out of the sixteen strategies, the most successful momentum strategy appears to be the one that is formed on the basis of the returns during the previous 9 months followed by a 3 months holding period for both developed and emerging markets. The 9-month/3-month global momentum strategy remains the most profitable strategy in emerging markets with higher returns of 2.41% per month (33% per annum) compared to 2.00% in the developed markets (26.8%). The comparison between developed and emerging markets shows that the developed markets appear to diminish the extent of the momentum profitability due to their greater level of information efficiency (see Cakici and Tan, 2014; Cakici et al., 2016). A further insight of Table 4 is that the negative momentum returns observed for 12 months originated in emerging markets.

The results from Tables 2 to 4 report a significant yield when international investors switch from one index to another. In general, we find that as the duration of the holding period increases, the momentum return tends to diminish which is consistent with Chan et al. (2000) although the latter study uses shorter observation and holding periods. We have identified an optimal strategy of 9 months/3 months under different specifications. We use international rather than country-specific indices or sampled stocks from different indices. By analysing momentum strategies based on stock market indices, this paper examines whether these strategies are useful for country selection and we report significant momentum returns given different country indices. Nevertheless, the success of the global momentum strategies demonstrates that trading on countries, rather than individual firms, yields significantly higher returns — three times higher. In line with the seminal empirical evidence outlined in this study, we assume that there is no restriction for investors in trading portfolios of stocks in individual markets worldwide. Investors who do not have positions in individual markets might not be able to implement such strategies because, in some markets, short-selling of equities is prohibited or stock index futures are not available (Menkhoff et al., 2012). These strategies are, however, useful to international money fund managers who need to reallocate funds across different markets.

To increase the power of the test we also perform similar analyses on momentum strategies with *overlapping* portfolios, where momentum in any month holds indices ranked in the previous J months. Table 5 refers to the whole sampled period (1969 to 2020) and we observe that practically all of the momentum strategies' returns are positive and the period that emerged as the most profitable in the previous analysis is also confirmed as part of the overlapping portfolios case: i.e. the 9-month/3-month period. The 9-month/3-month momentum strategy yields returns of 0.83% per month (10.4% per annum).[2]

[2] In unreported results with subsample periods, the most successful momentum generates 0.73% per month with a t-statistic of 4.18 and a p-value of 0.00 in established markets, 0.71% per month with a t-statistic of 2.89, and a p-value of 0.00 during the globalization period, 0.77% per month with a t-statistic of 4.36 and a p-value of 0.00 in developed markets, and 0.17% per month with a t-statistic of 0.45 and a p-value of 0.65 in emerging markets. More importantly, the momentum strategies remain on average profitable with overlapping portfolios. The 9-month/3-month is the most profitable strategy. It generates a positive and significant payoff of 0.83% per month with a t-statistic of 4.07 and a p-value of 0.00. These evidences also indicate that the momentum strategies' returns are

Table 3
Momentum returns with established markets and globalization period.

Momentum returns established markets* 1969–2020						Momentum returns since globalization** 1994–2020				
J	K	3	6	9	12	K	3	6	9	12
3 Sell		0.45	0.44	0.31	0.35		0.00	0.06	0.09	0.14
3 Buy		0.61	0.67	0.73	0.63		0.20	0.32	0.49	0.40
3 Buy-Sell		0.16	0.22	0.42***	0.27**		0.19	0.27	0.40*	0.26
		(0.57)	(1.16)	(2.62)	(1.98)		(0.47)	(0.98)	(1.93)	(1.46)
Sig		0.56	0.25	0.00	0.05		0.64	0.33	0.06	0.14
6 Sell		0.31	0.30	0.11	0.30		0.11	0.14	0.05	0.33
6 Buy		0.58	0.78	0.55	0.56		−0.02	0.55	0.46	0.39
6 Buy-Sell		0.27	0.48	0.44*	0.25		−0.12	0.41	0.41*	0.06
		(0.69)	(1.6)	(1.73)	(1.14)		(−0.23)	(1.16)	(1.78)	(0.26)
Sig		0.494	0.11	0.09	0.25		0.82	0.25	0.08	0.80
9 Sell		−0.40	−0.00	0.27	0.32		−0.46	0.13	0.052	0.15
9 Buy		1.27	0.60	0.70	0.64		1.64	0.70	0.66	0.47
9 Buy-sell		1.68***	0.60*	0.42	0.32		2.10***	0.57	0.60	0.33
		(3.33)	(1.86)	(1.45)	(1.16)		(2.76)	(1.36)	(1.62)	(0.89)
Sig		0.00	0.07	0.15	0.25		0.00	0.18	0.11	0.38
12 Sell		0.52	0.53	0.11	0.32		1.33	0.51	−0.08	0.33
12 Buy		0.71	0.50	0.16	0.44		0.04	−0.01	−0.13	0.07
12 Buy-		0.19	−0.03	0.05	0.12		−1.29*	−0.53	−0.05	−0.26
Sell		(0.28)	(−0.07)	(0.14)	(0.33)		(−1.84)	(−0.96)	(−0.14)	(−0.63)
Sig		0.78	0.95	0.89	0.75		0.08	0.35	0.89	0.53

This table reports the momentum strategies' returns implemented on 18 established stock indices from 1969 to 2014 and 47 stock market indices 1994–2014. We form momentum portfolio (buy past winners and sell past losers) based on past performance of stock market indices. We use 3 different formations and holding periods. The winner portfolios, the loser portfolios, and the winner minus loser portfolios average monthly returns are reported. The t-statistic are reported in the brackets and the results are statistically significant for $p < 0.05$.
**Established Markets*: USA, Japan, United Kingdom, Australia, France, Germany, Italy, Canada, Hong Kong, Singapore, Spain, Switzerland, Belgium, Sweden, Austria, Netherlands, Norway, and Denmark.
**: All countries — see Table 1.

Table 4
Momentum return with non-overlapping portfolios developed and emerging markets.

Momentum return in developed countries* 1969–2020						Momentum in emerging countries** 1988–2020				
J	K=	3	6	9	12	K	3	6	9	12
3 Sell		0.38	0.32	0.24	0.31		0.60	0.71	0.28	0.52
3 Buy		0.50	0.57	0.69	0.63		−0.11	0.02	0.36	0.47
3 Buy-Sell		0.12	0.25	0.45***	0.32**		−0.71	−0.69	0.08	−0.04
		(0.42)	(1.27)	(2.71)	(2.23)		(−1.06)	(−1.60)	(0.22)	(−0.16)
Sig		0.67	0.21	0.00	0.03		0.29	0.11	0.83	0.87
6 Sell		0.39	0.26	0.08	0.24		0.17	0.67	0.74	1.06
6 Buy		0.60	0.76	0.62	0.61		0.29	0.76	0.92	0.65
6 Buy-Sell		0.22	0.50*	0.54**	0.37*		0.12	0.09	0.19	−0.42
		(0.56)	(1.81)	(2.27)	(1.81)		(0.12)	(0.15)	(0.40)	(−0.99)
Sig		0.57	0.07	0.02	0.07		0.91	0.88	0.69	0.32
9 Sell		−0.69	−0.05	0.16	0.25		−0.89	0.73	0.54	0.63
9 Buy		1.30	0.58	0.74	0.67		1.53	0.80	0.91	0.53
9 Buy-sell		2.00***	0.63*	0.58*	0.41		2.41*	0.07	0.37	−0.10
		(4.30)	(1.86)	(1.85)	(1.42)		(1.72)	(0.09)	(0.63)	(−0.16)
Sig		0.00	0.07	0.07	0.16		0.09	0.93	0.53	0.87
12 Sell		0.59	0.24	−0.13	0.18		2.97	1.99	1.20	1.16
12 Buy		0.83	0.60	0.24	0.52		0.48	0.33	0.18	0.13
12 Buy-Sell		0.24	0.36	0.37	0.34		−2.49**	−1.66*	−1.02	−1.03
		(0.35)	(0.70)	(0.95)	(0.92)		(−2.14)	(−1.77)	(−1.18)	(−1.19)
Sig		0.73	0.48	0.35	0.36		0.04	0.09	0.25	0.24

This table reports the momentum strategies' returns with non-overlapping portfolio implemented on Developed and emerging stock market indices. We form Momentum portfolio (buy past winners and sell past losers) based on past performance of stock market indices. We use 3 different formation and 3 different holding periods. The winner portfolios, the loser portfolios, and the winner minus loser portfolios average monthly returns are reported. The t-statistic are reported in the brackets and the results are statistically significant for $p < 0.05$.
**Developed Markets*: USA, Japan, UK, Australia, France, Germany, Italy, Canada, Hong Kong, Singapore, Spain, Switzerland, Belgium, Sweden, Austria, Ireland, Netherlands, New Zealand, Norway, Portugal, Denmark, Finland, and Israel.
**Emerging Markets*: China, Brazil, India, Korea, Russia, Turkey, Indonesia, South Africa, Mexico, Taiwan, Thailand, Argentina, Malaysia, Chile, Colombia, Egypt, Poland, CZECH Rep, Hungary, Pakistan, SRI Lanka, Morocco, Peru, and Jordan.

positive and significant with all the holding periods. However, it is important to reiterate that on average the returns are lower but highly significant. Even more interesting, this study did not find evidence of consistent return continuation among countries' indices when using overlapping portfolios.

As it is evident, the 9-month formation period followed by a 3-month holding period seems to be the most profitable strategy irrespective of the market state and time horizon as the Tables 2, 3, 4, and 5 testify. This is in line with Jegadeesh and Titman (1993, 2001) as well as Moskowitz et al. (2012), who establish

the profitability of the momentum strategy in the short run. Furthermore, as He and Li (2015) show, momentum profitability is closely related to the market state and time horizon. The momentum strategy is profitable when the time horizon is short and unprofitable when the time horizon is long. Another possible explanation for the global momentum is that when dealing with several countries' indices, unexpected volatility is likely to reduce long-term growth and investment benefits as it is stated by Rafferty (2005).

4.2. Momentum strategies in different market phases

The main contribution of this section is the assessment of the stability of momentum strategies when it comes to generating profits in different market states. Such findings link up the relatively small body of literature that suggests the failure of various investment strategies during a financial crisis. Siganos and Chelley-Steeley (2006) report that the profitability of the momentum strategy is positively affected by the prolonged prevalence of the bear market state. Daniel and Moskowitz (2016) suggested several periods, known as momentum crashes, where the momentum strategy works badly, generating huge losses for the investors. Antoniou et al. (2013), Hanauer (2014), Hao et al. (2018), Kent and Moskowitz (2016), Kim and Suh (2018), and Yu (2012) report that momentum strategies yield positive results when the US market is growing or stay stable respectively. On the other hand, Griffin et al. (2003) argue that momentum profits are positive in both good and bad states, incompatible with the view that momentum is a reward for priced business cycle risk. Hence, different market phases are associated with different returns and the momentum strategies will not be an exception.

Table 6 attempts to shed further light on the performance of momentum under different market states. Based on our findings that the 9-months/3-month period emerges consistently and significantly as the optimal one, we apply this formation/holding strategy in different market phases for the full sample of indices. Table 6 shows on average that the positive returns outnumber the negative ones lending a further boost in favour of the momentum strategy. Table 6 shows that the first negative return occurs in period 7 (March 1975), where the return was −1.89%, then in period 12 (December 1978), where the return was −1.21%. Since 1969, global equity markets have experienced a relatively strong upward trend and therefore, much of the momentum strategy returns are positive, 49 out of 66 periods although the period with negative returns became more common after the great financial crisis. The negative momentum payoffs are strongly associated with the post-stock market events including the secondary banking crisis of 1973–1975, the Latin American debt crisis that originated at the end of the 1970s to 1982, the Japanese asset price bubble from 1986–2003, the black Monday 1987, the European currency crisis 1992–1993, the Asian financial crisis during 1997–98, the Russian financial crisis 1998, the burst of the technology bubble in 2000, and the 2008 financial crisis as well as the sovereign debt crisis and the subsequent slow recovery which called for state interventions in early and mid-2010.

After the initial negative returns, the momentum strategy mainly produces positive returns which are interrupted by certain negative performances. Interestingly the momentum strategy produced positive returns straight after the financial crisis in 2008. The momentum returns are observed to be positively high during the pre-crisis period and then, turn negative during the crisis period, and again reverse to generate high momentum returns during the post-crisis period although this trend appears to diminish towards the end of the 2010s. Momentum strategies tend to be linked with positive returns in good market states when the expected market risk premium is low and vice versa.

Table 5
Momentum returns with overlapping portfolios: Full sample.

J	K=	3	6	9	12
3 Sell		0.39	0.36	0.26	0.34
3 Buy		0.73	0.79	0.89	0.76
3 Buy-Sell		0.34*	0.43***	0.64***	0.42***
		(1.77)	(3.40)	(6.02)	(4.74)
Sig		0.08	0.00	0.00	0.0
6 Sell		0.25	0.15	0.12	0.30
6 Buy		0.79	0.92	0.85	0.68
6 Buy-Sell		0.55***	0.77***	0.73***	0.37***
		(2.73)	(5.61)	(6.69)	(4.10)
Sig		0.00	0.00	0.00	0.00
9 Sell		0.11	0.04	0.15	0.33
9 Buy		0.95	0.88	0.78	0.62
9 Buy-sell		0.83***	0.84***	0.63***	0.29***
		(4.07)	(5.84)	(5.44)	(2.98)
Sig		0.00	0.00	0.00	0.00
12 Sell		0.13	0.21	0.30	0.45
12 Buy		0.84	0.67	0.55	0.43
12 Buy-Sell		0.71***	0.45***	0.25**	−0.02
		(3.29)	(3.13)	(2.14)	(−0.16)
Sig		0.00	0.00	0.03	0.87

This table reports the momentum strategies' returns with overlapping portfolio implemented on 47 stock market indices. The sample is from 1969 to 2014. I form momentum portfolio (buy past winners and sell past losers) based on past performance of stock market indices. I use 4 different formations and holding periods. The winner portfolios, the loser portfolios, and the winner minus loser portfolios average monthly returns are reported. The t-statistic are reported in the brackets and the results are statistically significant for $p < 0.05$.

Nevertheless, the reversal of the negative returns appears to take place quite quickly and fast.

The negative returns recorded over the study period are on average associated with international stock market crisis dates and indicate that the momentum strategy does not pick stocks that have high unconditional expected returns, as the selected indices are conditional to the information available at the time of the trade. The observed pattern of initially positive momentum returns followed by negative returns or the switch between positive and negative momentum payoffs at different time point also suggests that global indices' changes after the formation period are not permanent over time and demonstrate how major global shocks in recent history affect the momentum performances (Jegadeesh and Titman, 1993).

4.3. Momentum returns following bear and bull markets

Table 6 links the momentum strategies with certain economic and financial events that took place and had an effect on the market. Table 7 takes a different look by examining the link between momentum and market phases rather than events. We identify bear and bull phases in global markets using data from the MSCI World Indices and apply the 9-months (formation)/3-month (holding) strategy as identified by our previous analysis. Having identified 22 and 44 bear and bull phases respectively over the sampled period which lasted 9 months and armed with our 9-month/3-month strategy, we can see that the momentum profitability holds on average during bear and bull phases. If the bear phases last 9 months, our momentum returns during the follow-up 3-month holding period exhibits an average positive return of 2.17% whereas the equivalent return for the bull markets' phase is 2.72% on average. Our investigation draws upon the well-established momentum findings and momentum "crashes" after prolonged bear periods (during the Great Crash and Great Financial Crisis) having individual stocks as a showcase (see Barroso and Santa-Clara, 2016; Daniel and Moskowitz, 2016; Jegadeesh and Titman, 1993) but our indices-based strategy

Table 6
Performance of momentum strategies in event time.

(Period) Date	Monthly return	(Period) Date	Monthly return	(Period) Date	Monthly return
(1) 30/09/1970	4.08	(23) 31/12/1987	4.47	(45) 29/09/2004	−1.66
(2) 30/06/1971	7.28	(24) 31/12/1987	4.18	(46) 29/06/2005	6.26
(3) 31/03/1972	0.46	(25) 30/09/1988	−6.37	(47) 31/03/2006	5.23
(4) 29/12/1972	7.66	(26) 30/06/1989	11.32	(48) 31/12/2006	1.48
(5) 28/09/1973	6.09	(27) 30/03/1990	−6.03	(49) 20/09/2007	2.13
(6) 28/06/1974	5.32	(28) 31/12/1990	6.84	(50) 30/06/2008	5.07
(7) 31/03/1975	−1.89	(29) 30/09/1991	0.45	(51) 31/03/2009	3.05
(8) 31/12/1975	7.83	(30) 30/06/1992	2.80	(52) 31/12/2009	−3.87
(9) 30/09/1976	2.45	(31) 31/03/1993	2.87	(53) 28/09/2010	3.36
(10) 30/06/1977	9.26	(32) 31/12/1993	−13.99	(54) 30/06/2011	3.39
(11) 31/03/1978	0.41	(33) 30/09/1994	5.19	(55) 29/03/2012	−0.31
(12) 29/12/1978	−1.21	(34) 30/06/1995	0.74	(56) 31/12/2012	−1.08
(13) 28/09/1979	5.25	(35) 29/03/1996	−6.41	(57) 29/09/2013	−0.90
(14) 30/06/1980	1.89	(36) 31/12/1996	6.99	(58) 30/06/2014	2.46
(15) 31/03/1981	6.78	(37) 30/09/1997	16.39	(59) 31/03/2015	−0.45
(16) 31/12/1981	−3.14	(38) 30/06/1998	5.57	(60) 31/12/2015	−2.39
(17) 30/09/1982	3.85	(39) 31/03/1999	10.67	(61) 29/09/2016	1.41
(18) 30/06/1983	10.79	(40) 31/12/1999	1.07	(62) 30/06/2017	0.58
(19) 30/03/1984	0.51	(41) 29/09/2000	−1.75	(63) 31/03/2018	0.44
(20) 31/12/1984	2.98	(42) 29/06/2001	6.15	(64) 31/12/2018	−1.44
(21) 30/09/1985	10.18	(43) 29/03/2002	6.13	(65) 29/09/2019	0.18
(22) 30/06/1986	2.57	(44) 31/12/2002	1.34	(66) 30/06/2019	−3.76

This table reports the average momentum returns, winner minus loser, portfolio in each 3 month following the formation period (9) with the exception of the first 9month of the sample. The sample is from December 1969 to January 202020.

Table 7
Description of the bear and bull phases 1969–2020.

	No of bear/bull periods	% of total periods	Average market monthly performance	9-month/3-month performance after bear/bull
Panel A: Bear phase				
3 months	70	35.00		
6 months	29	29.59		
9 months	22	40.00	−1.14	2.17
12 months	11	22.00		
Panel B: Bull phase				
3 months	130	65.00		
6 months	69	70.41		
9 months	44	60.00	1.33	2.72
12 months	39	78.00		

Table 7 shows the number of bear and bull market over the full sample period, their frequency, market performances in different bear and bull horizon (3, 6, 9, 12) between December 1969 and January 2014, and the average momentum profits accomplished after bull and bear markets with the optimum strategy (9-month/3-month). The bear phase (Panel A) is the periods when the market returns (MSCI World Index) is negative for 3, 6, 9, and 12 months before the test period and the bull state (Panel B) is when the market return is positive for 3, 6, 9, and 12 months.

appears to produce positive returns even when the markets are operating in a bear state. Note that the momentum strategy of 9-months/3-months formation and holding period respectively outperforms both states of the markets and remarkably the bear state (−1.14% per month). These observations suggest that the momentum strategy generates superior gains when the market rises quickly in the bull and/or falls slowly in the bear phase as the high market performance indicates a high and positive change in Indices prices and inversely. Therefore, a slow recovery could be seen as bad news for momentum investors (and we have some evidence following the 2008 financial crisis — see Table 6) while the reverse is not necessarily good news for global momentum investors comparatively. These results are in line with Gonzalez et al. (2005) that suggested that investors can achieve superior returns by adopting the momentum strategy after a bear phase, but this study reiterates that the return on the momentum strategy will depend upon the speed of the rising and the falling market phases. It is not uncommon for negative market states to produce positive momentum results, similar results are also reported by Griffin et al. (2005) although they focus on stocks worldwide. They report that during periods of negative returns across the markets momentum returns are positive in 35 out of 40 markets worldwide from 1975 to 1995.

4.4. Global risks impact on momentum strategy payoff

Table 8 reports the results of the momentum portfolio return (9-month/3-month) conditional upon global risk factors. Table 8 also reveals results for the arbitrage portfolio formed by buying the highest-ranked equities (winners) and selling the lowest-ranked equities (the losers). We begin the estimation with idiosyncratic risk factors, and then include market state and macroeconomic factors consecutively. If the momentum strategies' returns depend on the set of factors described by the models, then the intercept of the models, alpha *(a)*, should not be significantly different from zero.

In Table 8, Models 1, 2, and 3 report that the momentum effect persists (intercept $\alpha_1 > 0$ and significant) even when certain factors are accounted for and even after considering consecutively the idiosyncratic risks and the market state factors. The explanatory power of the Table 8 reports that the momentum effect persists (intercept $\alpha_1 > 0$ and significant). In Models 1, 2, and 3, the intercept remains positive and significant after considering consecutively the idiosyncratic risks, market state factors, and macroeconomic risks. This indicates that the explanatory power of the momentum returns does not reside with the factors employed by Models 1, 2, and 3, and the momentum strategy is able to deliver positive results from the outset without the need to rely heavily on the risk factors. In addition, when all

Table 8
Effect of global risk factor on global momentum profit (9-months/3-months formation and holding period).

	Model 1	Model 2	Model 3	Model 4
α_1	0.01***	0.01*	0.010**	0.01
	(4.37)	(1.80)	(2.28)	(1.28)
	0.00	0.07	0.02	0.20
LIQ		0.00		−0.14
		(0.03)		(−0.15)
		0.97		0.88
DS		−0.01		0.00
		(−1.04)		(0.46)
		0.30		0.65
TS		−0.00 **		−0.01*
		(−0.25)		(−1.70)
		0.803		0.09
MKT		1.22		−1.01
		(1.39)		(−0.60)
		0.16		0.55
ΔOP			−0.00**	0.00**
			(2.52)	(2.55)
			0.01	0.01
WVOL			−0.00***	−0.00***
			(−3.06)	(−2.50)
			0.00	0.01
ΔIP			1.16**	1.67**
			(1.99)	(2.49)
			0.05	0.01
MKTRF	0.05	−1.13	−0.30**	0.76
	(0.41)	(−1.30)	(−2.13)	(0.46)
	0.68	0.20	0.03	0.64
SMB	−0.23	−0.04	−0.30	−0.06
	(−1.15)	(−0.20)	(−1.44)	(−0.26)
	0.25	0.85	0.15	0.79
HML	−0.08	−0.18	−0.30	−0.16
	(−0.48)	(−0.88)	(−1.37)	(−0.79)
	0.63	0.381	0.17	0.43

This table reports the regression result based on model (7–10). Where: the dependent variable are the returns of the momentum portfolio at time t and regressed on, HML_t (high minus low) the return to portfolios that is long on high book-to-market stocks and short on low book-to-market stocks in US, SMB_t (Small minus big) returns to long-short portfolios constructed using size in US market, the LIQ_{t-1} the liquidity factor at time t−1, DS_{t-1} the Default spread at time t−1, TS_{t-1} the Term spread at time t−1, MKT_{t-1} the MSCI world indices return t−1, ΔOP_{t-1}, the change on monthly Oil price at time t−1, $WVOL_{t-1}$ is the market volatility at time t−1, and ΔIP_{t-1} is the change on monthly value of the US Industrial production at time t−1. I estimate the parameters using the interactive version of the generalized method of moment's estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors based on Bartlett kernel (Newey-West) with 1 lags. These regressions are based on model 1 to 4 specification. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

sets of explanatory variables are included (Table 8 Models 4) the intercept remains positive and significant. However, it appears that industrial production economically as well as statistically affects the momentum return with a coefficient estimate of 1.16 and a t-statistic of 1.99 when we control exclusively for macroeconomics risks. The impact of industrial production is, even more pronounce when we control jointly for idiosyncratic risks, market risks, and macroeconomics risks with a coefficient of 2.67 and a t-statistic of 2.49 (Table 8 Models 4).

Table 9 expands on the findings of Table 8 and provides a further investigation of momentum returns conditional upon certain risk and idiosyncratic factors. Table 8 Model 4 implies that when the full set of explanatory factors is used, the 9-month (formation)/3-month (holding) period renders the momentum returns statistically insignificant although the alpha coefficient remains positive. Given the importance of the 9-month formation period, we hold the latter constant and we vary the holding period at intervals of 3 months, formulating four different momentum strategies (9-month/3-month, 9-month/6-month, 9-month/9-month, and 9-month/12-month). We still have some evidence of significant momentum effects for the 9-month/6-month, 9-month/9-month period even when the full set of factors is considered. However, this effect dissipates with the 9-month/12-month momentum strategy and emphasizes the importance of having the right holding period. Furthermore, based on the sign of the coefficients, it appears that industrial production contributes positively to explaining the momentum abnormal in both the 9-month and 12-month holding periods, and the term spread and the market volatility tend to act as detrimental influences on momentum abnormal return. These findings indicate that macroeconomic risks factor have a significant impact on momentum return and this impact is associated with a specific holding period such as the 3-month, defined as the optimum holding period. This is consistent with Chordia and Shivakumar (2002) study that suggests that momentum strategies' profits are explained by common macroeconomic variables, implying that the profitability of the global momentum strategy could be due to variations in common macroeconomic factors and presumably changes in risks.

4.5. Business cycle role on momentum profits

We examine the joint effect of idiosyncratic risk factors, market states, and macroeconomic risk factors on momentum payoff during different phases of the business cycle. We follow Avramov et al. (2016) approach and we take into consideration expansionary and contractionary periods. To ensure consistency, we replicate the same models as in Table 8 and the momentum strategy remains unchanged (9-month/3-month). Table 10 Model 1 indicates that our momentum strategy produces significant abnormal returns of 0.01% per month having taken the idiosyncratic risk into consideration. Nevertheless, the momentum returns become insignificant when we account for market states and macroeconomic risks factors in Table 10 Models 2 through 4. It appears that during expansionary periods industrial production turns out to explain a significant proportion of the variation in momentum returns, rendering the alpha coefficient very small and insignificant in Models 3 and 4. Table 10 also verifies the findings of Table 8 when the full set of factors is considered, rendering the alpha coefficient insignificant. The default spread also contributes significantly in explaining the momentum profit with a positive and significant coefficient of 0.02 which falls behind, compared to the influence exhibited by the effect of industrial production which remains positive and significant with an accompanying sizeable coefficient as in Table 8. We can therefore confirm at this stage that momentum investors will improve their profit while considering the market state and mainly macroeconomic factors in expansionary periods.

Table 11 replicates the analysis discussed in Table 10, but now we turn our attention to momentum strategies during contraction periods. Table 11 Model 1 indicates that the momentum strategy yields a significant abnormal return of −0.016% per month with, a t-statistic of −3.08 after adjusting for idiosyncratic risk. This strongly supports the avowal that Fama–French's three-factor model is well specified for momentum portfolio and does not explain the momentum profit, at least for the short to medium term, in line with Fama and French's (1993). The subsequent analysis that includes market and macroeconomics risks factors consecutively in Table 11 Models 2 to 3 indicate insignificant adjusted momentum returns. As of interest, the coefficient associated with the macroeconomic risk factors such as industrial

production becomes insignificant in concurrence with other risk factors. This effect persists with all risk factors included (Table 11 Model 4) and indicates that the change in momentum returns in periods of contraction cannot solely be explained by a change in industrial production. However, it is noticeable that individual variable impacts are not consistent over time. One possible interpretation of this result is that there is a strong relationship between change in market state and the global momentum. I, therefore, confirm at this stage that if the momentum profits are improved by taking extra risks; these risks do at least or partially derive from market state and macroeconomic factors, consistent with Chordia and Shivakumar's (2002, 2006) studies that suggested that the short-term return continuation are primarily a result of cross-sectional variation in the exposure of momentum portfolios to the earning-based factor.

However, during the contractionary period, we can sense the emergence of a couple of factors that play a significant role, namely market volatility followed by term spread. Unsurprisingly, the default spread and the MSCI returns tend to suppress the momentum returns and they seem to do it in a significant fashion as Model 2 and Model 4 insinuate. These findings contribute to the debate by reinforcing Chordia and Shivakumar's (2002) study that suggested that momentum strategies' profits are explained by common macroeconomic variables, implying that the profitability of the global momentum strategy could be due to variations in common macroeconomic factors and presumably change in global risks.

4.6. Back-testing and Fama and French, and Xue and Zhang five factors

In a supplementary analysis of the global momentum return implemented with the Fama and French five factors, we estimated the adjusted returns as the intercept from the extended models 7 to 10 that account for the robust-minus-weak (RMW) profitability and conservative-minus-aggressive (CMA) investment factors. The results in Table 8A, Models 1, 2, and 3 suggest that the momentum effect persists (intercept $\alpha 1 > 0$ and significant) even when certain factors are accounted for. This reiterates that the explanatory power of the momentum returns does not reside with the factors employed by Models 1, 2, and 3, and the momentum strategy is able to deliver positive results from the outset without the need to rely heavily on these risk factors. In addition, when all sets of explanatory variables are included (Table 8A, Model 4) the intercept remains positive and significant. Industrial production continues to affect the momentum return with a coefficient estimate of 0.79 and a t-statistic of 3.37 (highly significant) when we control jointly for idiosyncratic, market, and macroeconomics risks. More importantly, the robust-minus-weak (RMW) profitability and conservative-minus-aggressive (CMA) investment factors show little strength in explaining the momentum profit. These results are consistent with Chordia and Shivakumar's (2002) study that suggests that momentum strategies' profits are explained by common macroeconomic variables, implying that the profitability of the global momentum strategy could be due to variations in common macroeconomic factors and presumably change in risks.[3]

Furthermore, we examine in Table 8B the extent to which the global momentum profits can be explained by Hou et al. (2021) five-factor model with a view to examine whether these factors can subsume the Fama and French factors and capture the global momentum return. Specifically, we replicate the models estimated in Table 8A using the Treasury bill rates, the market excess returns, the size factor returns, the investment factor returns, the return on equity factor returns, and the expected growth factor returns in place of Fama and French five factors model. The results are shown in Table 8B.

Table 8A
A Effect of global risk factor on global momentum profit.

	Model 1	Model 2	Model 3	Model 4
α_3	0.01 ***	0.01**	0.01**	0.01
	(4.18)	(2.18)	(2.37)	(1.60)
	0.00	0.03	0.02	0.11
LIQ		0.04		−0.00
		(1.05)		(−0.13)
		0.29		0.89
DS		−0.00		0.00
		(−0.33)		(0.34)
		0.74		0.74
TS		−0.00		−0.00
		(−0.99)		(−1.69)
		0.32		0.09
MKT		−0.03		−0.07
		(−0.62)		(−1.62)
		0.54		0.10
ΔOP			0.00**	0.00***
			(2.66)	(3.22)
			0.01	0.00
WVOL			−0.00***	−0.00***
			(−4.12)	(−4.58)
			0.00	0.00
ΔIP			0.69***	0.79***
			(2.91)	(3.37)
			0.00	0.00
MKTRF	−0.00	−0.00	−0.00	−0.00
	(−0.33)	(−0.18)	(−1.21)	(−1.00)
	0.74	0.86	0.23	0.32
SMB	−0.00	−0.00	0.00	0.00
	(−0.11)	(−0.10)	(0.16)	(0.20)
	0.92	0.92	0.87	0.84
HML	−0.00**	−0.00***	−0.00***	−0.00***
	(−2.32)	(−2.85)	(−3.11)	(−3.46)
	0.02	0.00	0.002	0.00
RMW	0.00	0.00	0.00	0.00
	(0.06)	(0.37)	(0.71)	(0.83)
	0.94	0.71	0.48	0.41
CMA	0.00	0.00	0.00	0.00
	(1.35)	(1.91)	(1.60)	(1.89)
	0.18	0.06	0.11	0.06

This table reports the regression result based on model (7–10). Where: the dependent variable are the returns of the momentum portfolio at time t and regressed on, HML_t (high minus low) the return to portfolios that is long on high book-to-market stocks and short on low book-to-market stocks in US, SMB_t (Small minus big) returns to long-short portfolios constructed using size in US market, we extend these model by adding, robust-minus-weak profitability (RMW) and low-minus-high (conservative-minus-aggressive) investment (CMA) to examine the impact of the five factors model, the LIQ_{t-1} the liquidity factor at time t−1, DS_{t-1} the Default spread at time t−1, TS_{t-1} the Term spread at time t−1, MKT_{t-1} the MSCI world indices return t−1, ΔOP_{t-1}, the change on monthly Oil price at time t−1, $WVOL_{t-1}$ is the market volatility at time t−1, and ΔIP_{t-1} is the change on monthly value of the US Industrial production at time t−1. For robustness checked we also include the RMW (Robust Minus Weak) and the CMA (Conservative Minus Aggressive) of the Fama and French five factor model. I estimate the parameters using the interactive version of the generalized method of moment's estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors based on Bartlett kernel (Newey-West) with 1 lags. These regressions are based on model 1 to 4 specification. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

[3] In unreported regressions that used Fama and French regional factors, the explanatory power of the industrial production is more pronounced in developed markets with a coefficient of 0.40 and a t-statistic of 1.97 (significant). However, the picture is slightly different in emerging markets where the industrial production coefficient becomes negative (−0.19) but it is insignificant, pointing to the fact the global momentum is more influenced by the industrial production taking place in developed countries.

Table 8B
Effect of global risk factor on global momentum profit.

	Model 1	Model 2	Model 3	Model 4
α_3	0.01***	0.01 *	0.01	0.01
	(3.06)	(1.75)	(1.60)	(0.91)
	0.00	0.08	0.11	0.31
LIQ		0.04		0.00
		(1.18)		(0.03)
		0.24		0.97
DS		−0.00		0.00
		(−0.08)		(0.24)
		0.93		0.81
TS		−0.00		−0.00*
		(−1.26)		(−1.97)
		0.21		0.05
MKT		−0.02		−0.06
		(−0.50)		(−1.32)
		0.68		0.19
ΔOP			0.00**	0.00***
			(2.54)	(3.04)
			0.01	0.00
WVOL			−0.00	−0.00***
			(−3.54)	(−3.81)
			0.000	0.000
ΔIP			0.67**	0.79***
			(2.73)	(3.31)
			0.00	0.00
R_MKTq5	0.00	0.00	−0.00	−0.00
	(0.11)	(0.26)	(−0.66)	(−0.49)
	0.91	0.79	0.51	0.62
R_MEq5	0.00	0.00	0.00	0.00
	(0.49)	(0.47)	(0.31)	(0.40)
	0.63	0.64	0.75	0.69
R_IAq5	−0.00	−0.00	−0.00*	−0.00
	(−0.70)	(−0.15)	(−1.67)	(−1.30)
	0.48	0.89	0.09	0.20
R_ROEq5	0.00**	0.00**	0.00	0.00
	(2.13)	(2.21)	(1.51)	(1.30)
	0.03	0.03	0.13	0.19
R_EGq5	0.00	0.00	0.00	0.00
	(0.65)	(0.78)	(1.17)	(1.29)
	0.56	0.44	0.24	0.20

This table reports the regression result based on model (7)–(10). Where: the dependent variable are the returns of the momentum portfolio at time t and regressed on Xue and Zhang (HXZ) (2020, 2021) five factors. We estimate the parameters using the interactive version of the generalized method of moment's estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors based on Bartlett kernel (Newey-West) with 1 lag. These regressions are based on model 1 to 4 specification. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

Models 1 and 2 in Table 8B suggest that the momentum effect persists (intercept $\alpha 1 > 0$ and significant) when the Hou et al. (2021) factors are measured taking into consideration idiosyncratic risks and the market state factors respectively. This indicates that the explanatory power of the momentum returns does not reside with the factors employed by Hou et al. (2021) and the momentum strategy is able to deliver positive results from the outset without the need to rely heavily on these risks factors. When we consider the Macroeconomics factors in Model 3 and consecutively when the set of all the explanatory variables are included (Table 8B Models 4) the intercept becomes insignificant pointing to the strength of the macroeconomic risk factors in explaining the global momentum profit. In addition, it appears that industrial production economically as well as statistically affects the momentum return with a coefficient estimate of 0.79 and a t-statistic of 3.31. The impact of industrial production is slightly less pronounced when it is contrasted with the corresponding Fama and French five factors model.

Table 9
Global risk factor and global momentum profit varying holding period.

	Model 1	Model 2	Model 3	Model 4
	3-Month	6-Month	9-Month	12-Month
α_2	0.01	0.02**	0.02**	0.01
	(1.28)	(2.17)	(1.90)	(1.00)
	0.20	0.03	0.06	0.32
LIQ	−0.14	0.03	−0.01	−0.08
	(−0.15)	(0.45)	(−0.15)	(−1.60)
	0.88	0.65	0.88	0.11
DS	0.00	0.00	0.01**	0.01**
	(0.46)	(0.67)	(2.37)	(2.84)
	0.65	0.50	0.02	0.01
TS	−0.01*	−0.00*	−0.00**	−0.00**
	(−1.70)	(−1.81)	(−2.26)	(−2.03)
	0.09	0.07	0.02	0.04
MKT	−1.01	−0.92	−2.00**	−1.99**
	(−0.60)	(−0.84)	(−2.03)	(−2.56)
	0.55	0.40	0.04	0.01
ΔOP	0.00**	0.00	0.00	0.00
	(2.55)	(1.38)	(0.58)	(0.08)
	0.01	0.17	0.56	0.94
WVOL	−0.00**	−0.00**	−0.00***	−0.00**
	(−2.50)	(−2.68)	(−2.96)	(−2.23)
	0.01	0.01	0.00	0.03
ΔIP	1.67**	0.76	0.54	0.64*
	(2.49)	(1.60)	(1.35)	(1.88)
	0.01	0.11	0.18	0.06
MKTRF	0.76	0.77	1.82*	1.93**
	(0.46)	(0.71)	(1.89)	(2.54)
	0.64	0.48	0.06	0.01
SMB	−0.06	0.34**	0.20	0.21
	(−0.26)	(2.77)	(1.30)	(1.57)
	0.79	0.01	0.19	0.12
HML	−0.16	−0.40**	−0.50***	−0.40***
	(−0.79)	(−2.33)	(−3.49)	(−3.64)
	0.43	0.02	0.00	0.00

This table reports the regression result based on model (7)–(10). Where: the dependent variable R are the returns of the momentum portfolio (9-month/3, 6, 9, 12-month) at time t and regressed on, HML_t (high minus low) the return to portfolios that is long on high book-to-market stocks and short on low book-to-market stocks in US, the SMB_t (small minus big) returns to long-short portfolios constructed using size in US market, the LIQ_{t-1} the liquidity factor at time t−1, the DS_{t-1} the Default spread at time t−1, TS_{t-1} is the Term spread at time t−1, the MKT_{t-1} the MSCI world indices return at time t−1, ΔOP_{t-1}, the monthly Oil price at time t−1, $WVOL_{t-1}$ is the market volatility at time t−1, and ΔIP_{t-1} is the monthly value of the US Industrial production at time t−1. I estimate the parameters using the interactive version of the generalized method of moment's estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors based on Bartlett kernel (Newey-West) with 1 lags. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

4.7. Global momentum implementation

From a practical investment perspective, it is important to indicate that the global momentum strategies will be profitable after accounting for transaction costs. On average, the momentum strategies with non-overlapping portfolios are rebalanced every 9 months, which results in very low turnover. The optimum strategy generates a return of 2.53% per month or about 35% per annum. On the other hand, Berkowitz et al. (1988) and Jegadeesh and Titman (1993, p.77) estimate one-way transaction costs of 23 basis points for institutional investors suggesting a transaction cost of 0.5% per trade with a 6-month/6-month strategy that appears to be conservative. An estimated transaction cost of 0.95% per annum as indicated by trustworthy market outlets such as Morningstar (Miller, 2022) for the global strategy suggests that the momentum profit remains at 34.05% per annum and does not undermine the high profitability of this strategy.

Table 10
Effect of global risks on global momentum profit during Expansion period (9-months/3-months formation and holding period).

	Model 1	Model 2	Model 3	Model 4
α_3	0.01***	−0.01	0.00	0.00
	(4.60)	(−1.09)	(0.81)	(0.08)
	0.00	0.27	0.42	0.94
LIQ		0.25**		0.15
		(2.40)		(1.39)
		0.02		0.17
DS		0.02*		0.02**
		(2.82)		(2.01)
		0.05		0.04
TS		−0.00		−0.01**
		(−0.65)		(−1.99)
		0.51		0.05
MKT		1.75*		−0.07
		(1.80)		(−0.04)
		0.07		0.97
ΔOP			0.00**	0.00**
			(2.08)	(2.25)
			0.04	0.03
WVOL			−0.00**	−0.00**
			(−2.62)	(−2.27)
			0.01	0.02
ΔIP			1.91**	2.25**
			(2.30)	(2.59)
			0.02	0.01
MKTRF	−0.14	−1.90**	−0.14	−0.13
	(−0.88)	(−2.04)	(−0.83)	(−0.07)
	0.38	0.04	0.41	0.94
SMB	0.05	−0.06	0.04	0.02
	(0.24)	(−0.26)	(0.19)	(0.08)
	0.81	0.80	0.85	0.933
HML	−0.31*	−0.04	−0.18	0.03
	(−1.81)	(−0.27)	(−0.96)	(0.15)
	0.07	0.789	0.34	0.880

This table reports the regression results based on Eqs. (7)–(10) Where: the dependent variable *R* are the returns of the momentum portfolio during expansion periods at time t and regressed on, HML_t (high minus low) the return to portfolios that is long on high book-to-market stocks and short on low book-to-market stocks in US, SMB_t (small minus big) returns to long-short portfolios constructed using size in US market, the LIQ_{t-1} the liquidity factor at time t−1, DS_{t-1} the Default spread at time t−1, TS_{t-1} the Term spread at time t−1, MKT_{t-1} the MSCI world indices return at time t−1, ΔOP_{t-1}, the change on monthly Oil price at time t−1, $WVOL_{t-1}$ is the market volatility at time t−1, and ΔIP_{t-1} is the change on monthly value of the US Industrial production at time t−1. Regression are based on model 1 to 4 specification. I estimate the parameters using the interactive version of the generalized method of moments estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors with based on Bartlett kernel (Newey-West) with 1 lags. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

Table 11
Effect of global risks on global momentum profit during contraction (9-months/3-months formation and holding period).

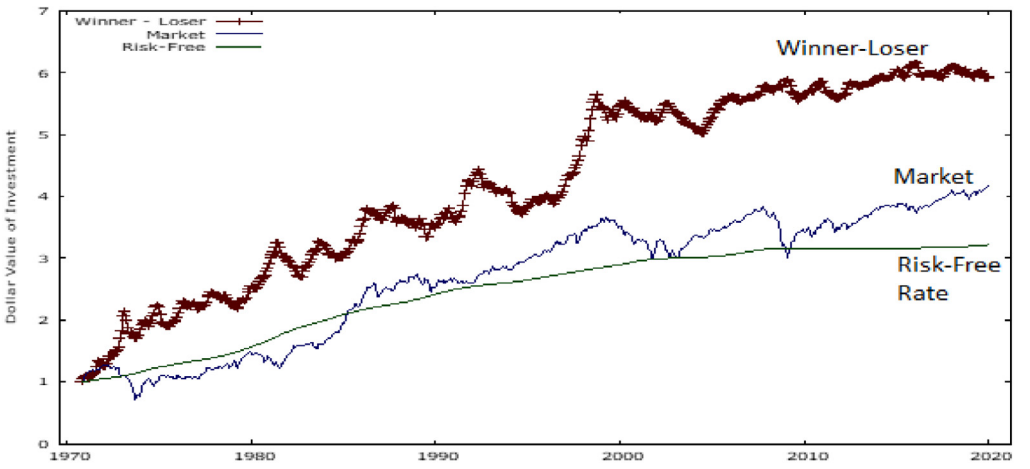
	Model 1	Model 2	Model 3	Model 4
α_4	−0.02*	0.03	0.05	0.11
	(−3.08)	(1.54)	(1.21)	(0.75)
	0.00	0.12	0.23	0.45
LIQ		−0.04*		0.48
		(−0.21)		(0.87)
		0.83		0.38
DS		−0.02		0.01
		(−0.82)		(0.12)
		0.41		0.90
TS		−0.01		−0.05
		(−0.70)		(−1.00)
		0.48		0.32
MKT		−1.50		−15.02
		(−0.49)		(−0.91)
		0.62		0.36
ΔOP			−0.00*	0.00
			(−1.66)	(0.62)
			0.09	0.53
WVOL			−0.00	−0.00
			(−0.71)	(−0.05)
			0.48	0.96
ΔIP			1.75	3.65
			(0.88)	(0.94)
			0.38	0.35
MKTRF	−0.43*	1.43	−1.78	17.30
	(−1.71)	(0.45)	(−1.41)	(1.00)
	0.09	0.65	0.16	0.32
SMB	−0.88 **	−0.24	0.44	0.18
	(−2.66)	(−0.55)	(0.38)	(0.11)
	0.01	0.58	0.70	0.92
HML	−0.06	−0.16	−0.87*	1.86
	(−0.37)	(−0.48)	(−1.74)	(0.72)
	0.71	0.629	0.08	0.47

This table reports the regression result based on model (7)–(10). Where: the dependent variable *R* are the returns of the momentum portfolio during contraction periods at time t and regressed on, HML_t (high minus low) the return to portfolios that is long on high book-to-market stocks and short on low book-to-market stocks in US, SMB_t (small minus big) returns to long-short portfolios constructed using size in US market, the LIQ_{t-1} the liquidity factor at time t−1, DS_{t-1} the Default spread at time t−1, TS_{t-1} the Term spread at time t−1, MKT_{t-1} the MSCI world indices return at time t−1, ΔOP_{t-1}, the change on monthly Oil price at time t−1, $WVOL_{t-1}$ is the market volatility at time t−1, and ΔIP_{t-1} is the change on monthly value of the US Industrial production at time t−1. Regressions are based on model 1 to 4 specification. I estimate the parameters using the interactive version of the generalized method of moment's estimation. I also account for time variation through time dummies. I request for heteroscedasticity and autocorrelation consistent standard errors based on Bartlett kernel (Newey-West) with 1 lags. I use the regression Alpha to measure the size of the abnormal return generated after adjusting for risk. The test statistics are reported in the parentheses and the p-value next. The results are * statistically significant at 10% for $p < 0.1$, ** statistically significant at 5% for $p < 0.05$ and ***statistically significant at 1% for $p < 0.01$.

Overall, the success of the global momentum strategies with non-overlapping portfolios presented in this section demonstrates that trading on indices performances worldwide might provide more evidence of market inefficiency when international investors invest in winners' countries and divest in losers' countries. This study relies on MSCI tradable indices and ETFs are easy to access, simple to use, and often have significantly lower transaction costs than many other types of investment funds (Miller, 2022). For example, a world equity ETF with a 5% turnover rate might incur transaction costs amounting to just 0.4 basis points per year.

4.8. Global momentum performance

Fig. 1 presents the momentum profit trend from 1969 to 2020 and provides an indication of a consistent increase in the momentum return in periods of good market states and a sharp decrease following a bad market state. In general, the good market states tend to predict good future momentum performances while the bad market states predict worsened momentum performances. Since 1969, global equity markets have experienced a strong upward trend and therefore, much of the momentum strategy returns are positive. The negative momentum payoffs are strongly associated with the post-stock market events including the secondary banking crisis of 1973–1975, the Latin American debt crisis that originated at the end of the 1970s to 1982, the Japanese asset price bubble from 1986–2003, the black Monday 1987, the European currency crisis 1992–1993, the Asian financial crisis during 1997–98, the Russian financial crisis in 1998, the burst of the technology bubble in 2000, the 2008 financial crisis, and Brexit in 2016. The consecutive decline in momentum payoffs after the first half of 2011 is also associated with the slow recovery following the 2008 financial crisis which called



Plotted are the dollar Value return on a dollar invested on the winner – loser, the risk-free asset and the market 1969 – 2020.

Fig. 1. Cumulative gain of investment 9-month/3-month 1969–2020. Plotted are the dollar value return on a dollar invested on the winner–loser, the risk-free asset and the market 1969–2020.

for unprecedented state intervention in many countries. In addition, Fig. 1 also highlights the persistent strength of the superior global momentum return relative to the risk-free rate and the stock market performance. More importantly, the observed pattern of initially rising momentum returns followed by a decline or switch between positive and negative momentum payoffs at different periods also suggests that global indices change after the formation period. This variability is illustrative of how major global shocks in recent history affect momentum performances (Jegadeesh and Titman, 1993).

5. Conclusion

We study global momentum strategies using data from 47 countries and for international investors that switch back and forth from one country to another in designing a global strategy. We find that international momentum strategies are consistently profitable during our research period starting in 1969 through 2020. The most successful momentum strategy selects stocks based on their previous performances over 9 months and then holds the portfolio for the next 3 months. This strategy yields 2.53% per month (35% per annum). Momentum returns vary considerably from one country group to another; 1.68% per month (22.13% per annum) in established markets, 2.00% (26.82% per annum) in developed countries, and 2.41% per month (33.08% per annum) in emerging markets. Furthermore, our initial findings that winning country investments are preferred over losing ones, do not support the hypothesis that information is immediately reflected in prices. The evidence suggests that prices respond gradually to new information supporting the under-reaction hypothesis (Chan et al., 1996; Savor, 2012).

After examining the best (9-month/3-month) momentum strategy in different market states, we find that there is a strong link between the rising market performances in the bull market and the momentum profits. The average momentum in the bull markets is 2.72% % per month while the bear market shows an average momentum return of 2.17%. The 9-month period provides the highest number of bull periods by definition (40% of the total number of periods) during the bear market by comparison to 3-month, 6-month, and 12-month. This indicates that the return(s) on the momentum strategy will also depend upon the length of the falling and the rising market phases.

We investigate if momentum profits remain significant when we account for different risk factors to identify if they are compensation for any particular risk. We find that global momentum strategies generate a risk-adjusted return of 0.010% per month when we control for idiosyncratic risks and 0.011% per month when we control for market state risks. However, alpha remains at 0.015% when we control for macroeconomic risks. Among others, world industrial production tends to contribute most in explaining the momentum returns when all risk factors are jointly accounted for. The impact of macroeconomic risk factors on the global momentum profit holds in emerging countries, developed countries, and established markets, and during the globalization phase of the world.

Our findings also echo that, industrial production contributes significantly in explaining the momentum return during contraction and expansionary periods, but the abnormal return varies from the contraction period (0.00%) to the expansionary period (0.011%) in line with Chordia and Shivakumar (2002) who suggested that variation in momentum payoffs reflects time-varying over the business cycle. These findings help deal with one of the fundamental issues in Finance, the Efficient Market Hypothesis which suggests that excess returns cannot be earned using investment strategies based on historical shared price or other historical data (Fama, 1970). As the results suggest if momentum investors carry higher macroeconomic risks such as Industrial Production and the State of the market they will earn extra returns.

To our knowledge, this is the first paper to show an easily implemented momentum strategy that switches back and forth country indices and generates extraordinarily high abnormal returns more than 2.53% per month. We also show these returns are compensation for global risks and for investors ready to take them. Further work would be recommended to study a longer period dating back to the nineteenth century or the Victorian Era characterized by the rapid change in economic development in almost every domain to verify if momentum is historically a compensation for carrying macro-economic risks. This era is shaped by advances in scientific, technological, and medical knowledge, and change in population growth and location, which deeply change countries' moods leading to optimism, economic boom, and prosperity. Balvers and Wu (2006), for example, suggest that technological progress in one country implies that the country has a competitive advantage that grows relatively fast as the

Table A.1

Symbol	Variable	Definition	Source	ID
ERM	Excess Return on the Market	ERM (or Rm-RF) is the excess return on the market. It is calculated as the value-weight return on all NYSE, AMEX, and NASDAQ stocks (from CRSP) minus the one-month Treasury bill rate from Ibbotson, Associates, Ken French's web site http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/ (Percent change)	WRDS	Library: ff File: Factors_monthly
MKTRF	Excess Return on the MSCI world market	MKTRF (or %Δ MSCIW-RF) is the excess return on the market. It is calculated as the value-weight return of the MSCI World index minus the one-month Treasury bill rate	Own calculation	
SMB	Small-Minus-Big Return	SMB (Small Minus Big) is the average return on the three small portfolios minus the average return on the three big portfolios, SMB =1/3 (Small Value + Small Neutral + Small Growth)−1/3 (Big Value + Big Neutral + Big Growth). SMB for July of year t to June of t+1 include all NYSE, AMEX, and NASDAQ stocks for which market equity data for December of t−1 and June of t, and (positive) book equity data for t−1, exists, from Ken French's web site. (Percent change)	WRDS	Library: ff File: Factors_monthly
HML	High-Minus-Low	Average return on the two value portfolios minus the average return on the two growth portfolios. HML = 1/2(small value + Big Value −1/2(small Growth + Big Growth), from Ken French's website). (percent change)	WRDS	Library: ff File: Factors_monthly
RF	Risk-Free	Rate of one Month Treasury Bill	WRDS	Library: ff File: Factors_monthly
	Fama and French Regional Factors Develop and Emerging Market	SMB is the equal-weight average of the returns on the three small stock portfolios for the region minus the average of the returns on the three big stock portfolios, SMB = 1/3 (Small Value + Small Neutral + Small Growth) – 1/3 (Big Value + Big Neutral + Big Growth). HML is the equal-weight average of the returns for the two high B/M portfolios for a region minus the average of the returns for the two low B/M portfolios,	Online available from https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_3developed.html	Data library

Table A.2

Symbol	Variable	Definition	Source	ID
LIQ	liquidity factor (PS_LEVEL)	Pastor–Stambaugh Level of Aggregate Liquidity (non-traded factor). PS_LEVEL basically correspond to equation numbers (5) in the published (JPE, 2003) version of the paper "Liquidity Risk and Expected Stock Returns".	WRDS	Library: ff Files: Liq_ps
TS	The term spread	Difference between the average yield of Treasury bonds with more than 20 years to maturity and the average yield of T-bills that mature in three months. TS (t) = LGB (t) – TB (t−1), LGB (t) is the return on a portfolio of long-term government bonds obtained from WRSD; TB (t−1) is the lag average yield of the T-bills that matures in three months.	Own calculation	
LGB		LGB is the Index Level Associate with 20-year bond return (B20) obtained from the Center for Research in securities prices data files	WRDS	Library: Crspa File: cti
TB		TB is Index Level Associate with the 90 Day bill returns obtained from the Center for Research in securities prices data files	WRDS	Library: Crspa File: cti
DS	Default Spread	Difference between the average yield of bonds rated BAA by Moody's and the average yield of bonds with a Moody's rating of AAA. based upon the Federal Reserve Board's H.15 release that contains selected interest rate for U.S.	Own calculation	

(*continued on next page*)

Table A.2 (*continued*).

Symbol	Variable	Definition	Source	ID
AAA	Moody's Aaa Bon rate	Average yield of bonds rated AAA, based upon the Federal Reserve Board's H.15 release that contains selected interest rate for U.S.	WRDS	Library: frb File: rates
BAA	Moody's Baa Bon rate	Average yield of bonds with a Moody's rating of BAA, based upon the Federal Reserve Board's H.15 release that contains selected interest rate for U.S.	WRDS	Library: frb File: rates
MSCI World	Market Indices	Value-weighted MSCI world index prices level	DataStream	Mnemonic: TRVS1MB Code: S31ovJ
ΔMKT	Market return	$\Delta MKTt = \ln(Pt) - \ln(Pt-1)$, where Pt is the MSCI World index price at time t and Pt−1 is the world index prices at time t−1	Own calculation	

Table A.3

Symbol	Variable	Definition or Source	Source	ID
OP	Oil price	Producer price index crude petroleum series obtain from the bureau of Labour Statistics	Bureau of Labour Statistics U.S. department of Labour Online available from http://data.bls.gov/pdq/SurveyOutputServlet	Crude petroleum-WPU0561
VOL	Historical Volatility	Monthly standard deviation of daily returns (22 trading days per month).	Own calculation	
IP	Industrial Production	Total Industrial Production and Manufacturing Production index (SDDS+)	Online available from http://www.federalreserve.gov/releases/g17/download.htm	AIP_SA_IX
ΔIP	Percent change Industrial production	Monthly differences in the logarithm of the industrial production indexes $\Delta IP = \ln(IPt) - \ln(IPt-1)$, IPt is the industrial production index level at time t and IPt−1 is the industrial production index level at time t−1	Own calculation	
ΔOP	Percent change Oil Price	Monthly differences in the logarithm of the producer price index crude petroleum series (obtain from the bureau of Labour Statistics, U.S. department of Labour,) $\Delta OP = \ln(OPt) - \ln(OPt-1)$, OPt is the producer price index level at time t and OPt−1 is the producer price index level at time t−1	Own calculation	

Table A.4

Symbol	Variable	Definition	Source	ID
NBER	Business cycle	Business cycle series is an interpretation of US Business Cycle Expansions and Contractions data provided by The National Bureau of Economic Research (NBER) is composed of dummy variables that represent periods of expansion (1) and recession (0).	Online available from http://www.nber.org/cycles/cyclesmain.html	
CC	Currency crisis	Annual depreciation versus the US dollar (or relevant anchor currency, historically the UK pound, the French franc, or German DM and presently the euro) of 15 percent or more. It is composed of dummy variables that represent periods of depreciation (1) and normal period (0)	Online available from http://www.carmenreinhart.com/data/browse-by-topic/topics/7/	
SMC	Stock market crisis	Stock market crisis series is based on the bestselling *This Time Is Different: Eight Centuries of Financial Folly* by Carmen M. Reinhart & Kenneth S. Rogoff. Using data developed by Reinhart and Rogoff. It maps the cyclical history of financial crisis since 1810. It is composed of dummy variables that represent periods of crashes (1) and normal periods (0)	Online available from http://www.carmenreinhart.com/data/browse-by-topic/topics/7/	
BC	Banking crisis	Banking crisis are mark by two types of events: (1) bank runs that lead to the closure, merging, or takeover by the public sector of one or more financial institutions; (2) if there are no runs, the closure, merging, takeover or large-scale government assistance of an important financial institutions. It is composed of dummy variables that represent periods of crisis (1) and normal period (0)	Online available from http://www.carmenreinhart.com/data/browse-by-topic/topics/7/	

technology is implemented leading to momentum profitability for this country. When the technology is imitated in other countries, and the production levels converge, that could cause a reversion in equity prices worldwide in earlier periods of history as well.

Appendix A. Price risk variable (Eq. (7))

See Table A.1. Definitions for Price Risk Variables.

Table A.5

Symbol	Variable	Definition	Source	ID
Euribo	Euribor	Euribor if the rate at which Euro interbank term deposits are offered by one prime bank to another prime bank within the EMU zone is calculated at 11:00 a.m. (CET) for spot value (T+2). The choice of banks quoting for Euribor is based on market criteria. These banks have been selected to ensure that the diversity of the euro money market is adequately reflected. Thereby making Euribor an efficient and representative benchmark.	Online available from https://www.emmi-benchmarks.eu/euribor-org/about-euribor.html	
Eonia	Eonia	Eonia (Euro Overnight Index Average) is the effective overnight reference rate for the euro. It is computed as a weighted average of all overnight unsecured lending transactions in the interbank market, undertaken in the European Union and European Free Trade Association (EFTA) countries. The European Central Bank is the calculation Agent for Eonia.	Online available from https://www.emmi-benchmarks.eu/euribor-eonia-org/about-eonia.html	
EOIS	EURIBOR: OIS Spread	It is the difference between the rate at which European banks lend to each other (EURIBOR) and the overnight 'risk free' swap rate (EONIA) among the same banks a 3-month period. EURIBOR (Euro Interbank Offered Rate) is an average of the rate each bank in the 43-member 'prime bank' panel reports that it would offer to the other banks. EONIA (Euro Overnight Index Average) is the average of swaps conducted between a 22-member panels at what each panel bank believes is the mid-market rate each day.	Own calculation	
TED	TED Spread	It is the difference between the LIBOR (London Interbank Offered Rate) and the 3 Month Treasury Bill.	Own calculation	
Libor	Libor	It is the average interbank interest rate at which a selection of banks on the London money market are prepared to Lend to one another. Libor comes in 7 maturities (from overnight to 12 months) and in 5 different currencies. The official Libor interest rates are announced once per working day at around 11:45 a.m. In the past, the BBA/IAE published LIBOR rates for 5 more currencies (Swedish hrona, Danish krone, Canadian dollar, and Australian dollar and New Zealand dollar) and 8 more maturities (2 weeks, 4, 5, 7, 8, 9, 10 and 11 months).	Online available from http://www.global-rates.com/interest-rates/libor/libor.aspx	

Appendix B. Market state variable (Eq. (8))

See Table A.2. Definitions for Market State Variables.

Appendix C. Economic variable (Eq. (9))

See Table A.3. Definitions for Economic Variables.

Appendix D. Crisis

See Table A.4. Definitions for Crisis Variables.

Appendix E. Other variables

See Table A.5. Definitions for other Variables.

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